

DIEM-A Challenge

Dataset

Train 74 performers (JP 40 · TW 34) → 7,992 clips
Test 18 performers (JP 9 · TW 9) → 1,944 clips

12 emotions

3 scenarios × 3 intensities | BVH·FBX·C3D + text | performer-disjoint LPO

25-joint skeleton · 64-frame clip

Pipeline

11 models · 4 inductive biases

GCN family — STGCN++ · CTR-GCN · ProtoGCN

Attention — SkateFormer

Hybrid / MLP — Conv1D+Tr · Region-Aware · KP-MLP

External pretrain — MotionBERT · C3D-stats · MAMP×2

**Logit-mean fusion
equal-weight 11-way**

12-class softmax → emotion

Outcomes

Performance
Macro-F1 36.94%
(STGCN++ 25.2%)

Explainability
kinematics · body parts
· saliency maps